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Development and Preliminary Evaluation of an Electric Kick-Scooter Platform with BLDC Drive Control, Battery Voltage–Current Logging, and Rider-Mass-Based Theoretical Range Estimation

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Abstract

Electric kick scooters are becoming popular for short-distance applications on campuses, in industrial facilities and in gated communities, but their electrical demand and integrated low-cost operational functions are not fully addressed. A 350 W brushless direct-current hub motor, a 36 V, 4.4 Ah lithium-ion battery pack and a Hall-sensor-based controller with 120° six-step trapezoidal commutation were developed. A battery voltage–current data logger, a preliminary rider-mass-based theoretical range-estimation approach and a mobile application prototype were also developed. The gate states and phase-terminal voltage responses were simulated in MATLAB/Simulink R2021b, while preliminary hardware testing confirmed motor operation. The maximum observed motor speed was 1753 rpm, and the minimum was 1632 rpm. During partial charging and load-based discharging, the battery voltage and current were recorded by the data logger. Average currents of 3.88, 7.40 and 8.30 A were recorded in three pilot trials involving rider masses of 60, 80 and 140 kg, respectively. Assuming nominal battery energy and an average speed of 15 km/h, the corresponding theoretical ranges were calculated as 17.01, 8.92 and 7.95 km. These results demonstrate battery voltage–current acquisition by the data logger and provide descriptive pilot observations across the rider-mass cases; they do not establish a validated predictive or causal relationship between rider mass and actual scooter range.

Keywords: electric kick scooter; brushless direct-current motor; six-step commutation; battery voltage-current logging; electric micro-mobility; rider-mass-based theoretical range estimation

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1. Introduction

Electric kick scooters are increasingly used for short-distance travel on university campuses, in industrial facilities, in gated communities, and in similar controlled environments because of their compactness, ease of deployment, and relatively low operating cost [1–4].

Their practical operation depends on the motor-drive system, battery subsystem, and user interface [5–7]. Previous studies have addressed these elements mainly as separate research problems. Corno et al. [8,9] developed electrically assisted kick scooters with kick detection and closed-loop speed control, whereas Asperti et al. [10] examined the vertical dynamics, rider comfort, and road-holding behaviour of a kick scooter. BLDC hub motors are widely used in light electric-mobility applications because of their compact structure, high torque density, and low maintenance requirements [11]. Hall-sensor-based six-step commutation remains a practical low-complexity control approach for such applications [12–14]. Electrical energy demand is also an important consideration. Previous studies have shown that rider mass, road gradient, acceleration, and stop–start operation can affect the energy consumption of electric vehicles and light electric-mobility systems [15–18]. Battery-related studies have considered internal resistance, temperature effects, and state-of-charge estimation [19], while IoT-based systems and mobile interfaces have been used for vehicle-status monitoring, diagnostics, and fleet operation [20–22].

Table 1 provides a feature-level comparison between the present study and representative published studies. Building on these individual contributions, the present study develops and preliminarily evaluates an electric kick-scooter prototype incorporating a Hall-sensor-based BLDC controller, battery voltage–current logging, a limited rider-mass energy comparison, and a prototype mobile interface. The controller and data logger were evaluated through subsystem tests, whereas the rider-mass investigation was limited to three pilot trials and theoretical energy-based range estimates.

Table 1. Feature-level comparison of the present work with representative studies.

Aspect	Corno et al. [8,9]	Asperti et al. [10]	EV/e-Bike Energy Studies [15–18]	Present Work
Vehicle type	Kick scooter	Kick scooter	EVs/e-bikes	Kick scooter
Primary focus	Kick detection and speed control	Vertical dynamics and rider comfort	Energy-consumption	Preliminary system-level prototype
Experimental evidence	Road testing	Test-bench testing	Simulation and experimental studies	Subsystem tests and three pilot trials
Rider-mass analysis	Not a primary focus	Considered in vehicle modelling	General payload effects	Three rider-mass pilot cases
Range information	Not reported	Not a primary focus	Model-based or measured-energy analysis	Theoretical energy-based estimate
Motor-control approach	Closed-loop speed control	Not a primary focus	Not a primary focus	Hall-sensor-based six-step BLDC control
Battery-related measurement	Not a primary focus	Not a primary focus	Limited or application-specific	Time-stamped voltage–current logging
Mobile connectivity	Not reported	Not reported	Limited	Prototype mobile interface

The comparison provides feature-level context and does not constitute a direct performance benchmark. Accordingly, this study contributes (i) a Hall-sensor-based BLDC motor controller using six-step, 120° trapezoidal commutation; (ii) a time-stamped battery voltage–current data logger; (iii) a preliminary comparison of three rider-mass pilot trials using average-current measurements and theoretical range calculations; and (iv) a prototype mobile interface for intended vehicle-status display and selected control functions. The developed system is intended as a preliminary prototype for evaluating drive control,

battery voltage–current logging, rider-mass-related electrical demand, and mobile-interface functions. The battery subsystem was limited to terminal-voltage and current acquisition and logging; state-of-charge and state-of-health estimation, battery modelling, parameter identification, capacity characterisation, and thermal analysis were outside the scope of this study. Similarly, the range values are theoretical estimates rather than validated predictions of actual travel distance.

2. System Design and Methodology

2.1. Overall System Architecture

The developed electric kick scooter consists of a 350 W, 6-inch brushless direct-current (BLDC) hub motor, a Hall-sensor-based motor controller, a 36 V, 4.4 Ah lithium-ion battery pack, a battery management system (BMS), user-control inputs and an Internet of Things (IoT)-based control and communication board. The overall system architecture is shown in Figure 1. The 10-series, 2-parallel (10S2P) battery pack supplies power to the motor controller through the BMS. The BMS provides overvoltage, undervoltage and overcurrent protection, as well as cell balancing. The motor controller features a six-switch inverter and includes current protection, pulse-width modulation (PWM) control and 120° trapezoidal commutation. The BLDC hub motor receives three-phase power from the motor controller, and the signals from the Hall sensors are used as feedback for the commutation process [23]. The PWM duty cycle depends on the throttle input and, hence, determines the average voltage applied to the motor. The electronic brake input supplies a brake or motor-disable signal. The horn, headlamp and left/right indicator switches are connected to the IoT control and communication board. The IoT board operates the horn, headlamp, tail lamp and direction indicators and facilitates Bluetooth communication. It also has a two-wire controller area network (CAN) interface to communicate with the motor controller. The organic light-emitting diode (OLED) display shows vehicle speed, battery voltage, riding mode and system-warning information. State-of-charge estimation was not implemented and validated in the present study.

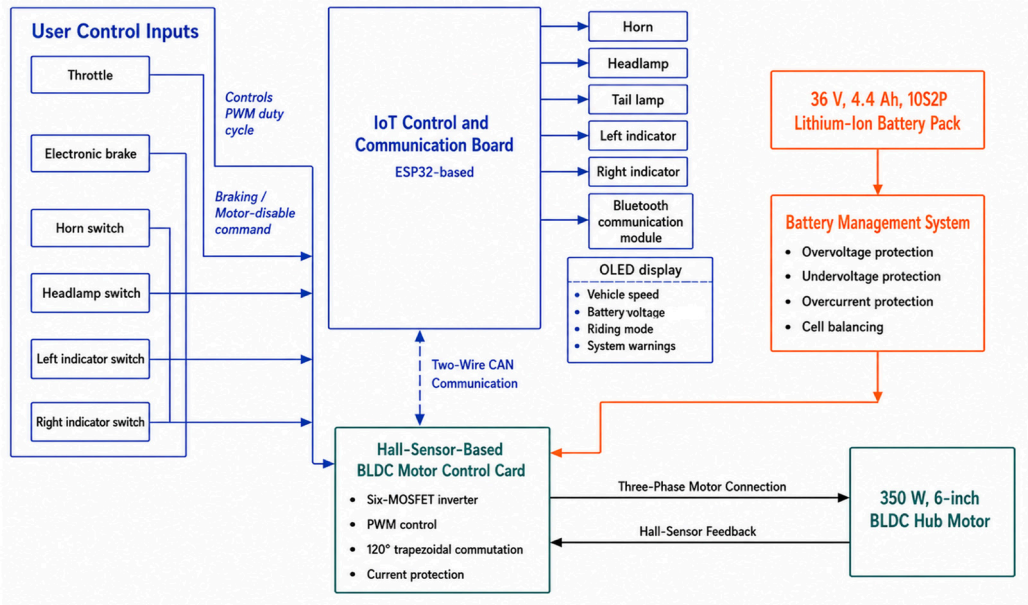


Figure 1. Overall architecture of the developed electric kick-scooter system. Blue indicates control, communication, and display; orange indicates the battery, BMS, and power supply; blue-green indicates the motor controller and motor. Black arrows show motor connections and Hall-sensor feedback.

2.2. BLDC Motor and Controller Design

2.2.1. Reduced-Order Motor Relation

A reduced-order BLDC motor relationship was used for controller development. With trapezoidal back-electromotive force and balanced three-phase operation, the electromagnetic torque is estimated by Equation (1) [24,25].

$$T_e \approx K_t i_{phase} \quad (1)$$

where T_e is the electromagnetic torque, K_t is the motor torque constant, and i_{phase} is the effective phase current that contributes to torque production. This relation is mainly used for the practical design of the controller and does not represent a detailed electromagnetic model of the BLDC motor. Instead, it describes the dominant current–torque behaviour of the motor.

2.2.2. Hall-Sensor-Based Commutation Strategy

The three Hall-sensor signals, H_a, H_b, and H_c, are used to implement 120° trapezoidal commutation by determining the rotor position. For each valid Hall state, the controller switches the corresponding upper-side and lower-side metal–oxide–semiconductor field-effect transistors (MOSFETs), energising two motor phases while keeping the third phase floating. The six-step switching sequence generates the rotating magnetic field required for continuous torque production. The switching states are shown in Table 2. Here, MOSFETs A1, B1, and C1 are the upper-side MOSFETs of phases A, B, and C, respectively, while MOSFETs A2, B2, and C2 are the corresponding lower-side MOSFETs. To verify the Hall-state-to-phase mapping, the motor wiring and Hall-state sequence were configured for the developed prototype.

Table 2. Hall-sensor-based six-step switching sequence.

Hall Sensor Input			Motor Phases			Switching Sequence					
Ha	**Hb**	**Hc**	**A**	**B**	**C**	**A1**	**A2**	**B1**	**B2**	**C1**	**C2**
1	1	0	OFF	+	-	0	0	1	0	0	1
0	1	0	-	+	OFF	0	1	1	0	0	0
0	1	1	-	OFF	+	0	1	0	0	1	0
0	0	1	OFF	-	+	0	0	0	1	1	0
1	0	1	+	-	OFF	1	0	0	1	0	0
1	0	0	+	OFF	-	1	0	0	0	0	1

For practical inverter operation, the upper-side and lower-side MOSFETs in the same inverter leg must not conduct simultaneously, because this condition can cause shoot-through and excessive current through the switching devices. Appropriate gate-signal interlocking and a short dead-time interval are therefore required when switching between complementary MOSFETs. Excessive dead time can also affect the effective phase voltage and switching performance. In the present controller design, the Hall-sensor signals determine the required commutation sector, while PWM is applied to the switching devices selected by the active commutation state. The PWM duty cycle is varied according to the throttle input to modulate the average voltage supplied to the energised motor phases and, consequently, the motor speed under the tested operating conditions. The implemented dead-time value and its effects on switching losses, torque ripple, and controller efficiency were not experimentally quantified in the present study.

2.2.3. Controller Hardware

The BLDC motor controller prototype was designed using a microcontroller, Hall-sensor input circuits, gate-driver stages, and a six-MOSFET three-phase inverter. The three Hall-sensor signals provide the microcontroller with information about the rotor position. Based on this information, the microcontroller selects the appropriate commutation state from the six-step sequence presented in Table 2. The throttle signal is used to modulate the PWM duty cycle applied to the active MOSFET pair, while the electronic-brake input sends a motor-disable or braking command. The control signals from the microcontroller are transmitted through the gate-driver stage to the upper and lower MOSFETs of the three inverter legs. The inverter supplies all three phases, A, B, and C, of the 48 V BLDC hub motor, while the Hall-sensor signals H_a, H_b, and H_c are fed back to the controller. Current protection and voltage regulation were also incorporated to ensure safe controller operation. The functional structure of the controller is shown in Figure 2. Under approximately constant load conditions, the motor speed mainly depends on the average voltage supplied to the motor through PWM, estimated by Equation (2) [26].

$$\omega_r \propto D V_{dc} \tag{2}$$

where ω_r is the rotor angular speed, D is the PWM duty ratio, and V_{dc} is the DC-link voltage.

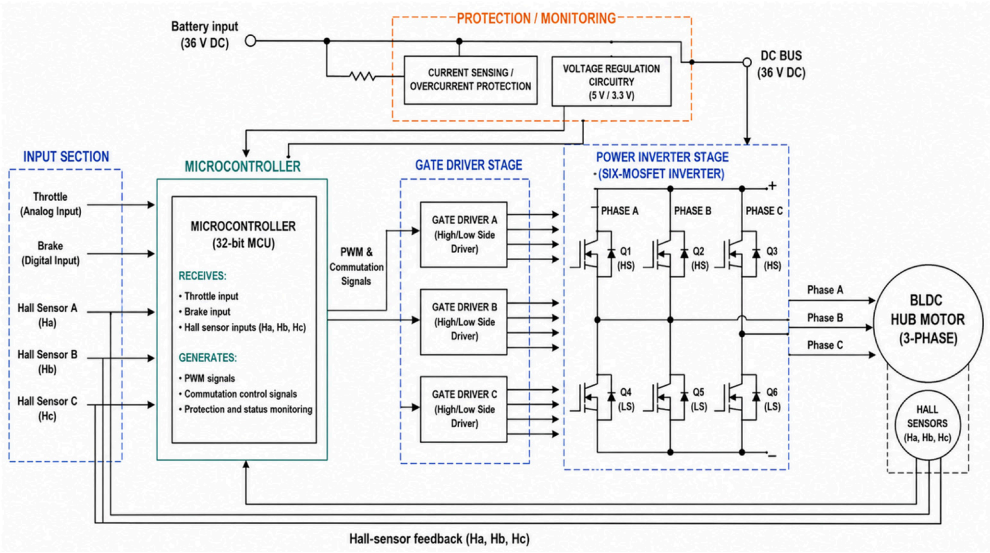


Figure 2. Functional architecture of the Hall-sensor-based BLDC motor controller [27].

2.2.4. Simulation of Switching Logic

The six-step inverter switching logic was simulated in MATLAB/Simulink before hardware implementation. The simulation model consisted of six switching-signal inputs applied to the upper- and lower-side devices of a three-phase inverter comprising six switching devices, as shown in Figure 3.

The three inverter outputs were connected to a balanced star-connected resistive load corresponding to phases A, B, and C, and voltage measurement blocks were used to measure the terminal-voltage response of each phase under a direct-current supply. The six gate states and the resulting three-phase voltage waveforms were observed using separate scope blocks. A discrete simulation step of 5×10^{-5} s was used to execute the model. The simulation verified the six-step switching sequence, the relative timing of the inverter gate signals, and the generation of positive, zero, and negative phase-voltage states. The MATLAB/Simulink model was developed solely to verify the six-step inverter

switching sequence, gate-signal timing, and corresponding phase-terminal voltage states. It was not intended to represent a complete BLDC motor model. Therefore, electromagnetic characteristics, back-electromotive-force (BEMF) generation, electromagnetic torque, motor inertia, mechanical load, and speed dynamics were not simulated. The simulation results should consequently be interpreted only as verification of the inverter switching logic and voltage response.

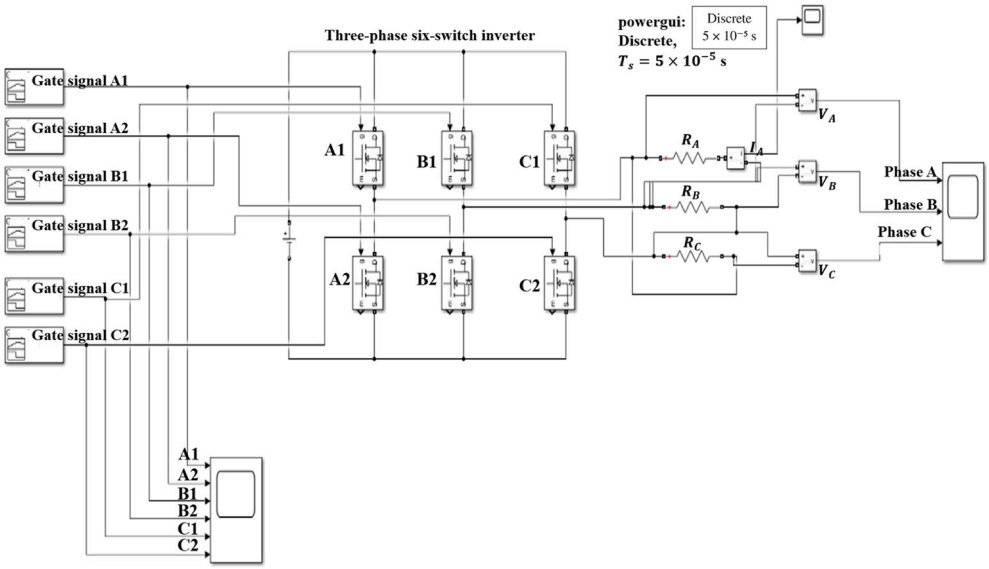


Figure 3. MATLAB/Simulink model used to verify the six-step inverter switching sequence and voltage response.

2.3. Real-Time Battery Data Logger Design

A prototype battery voltage–current data logger was developed to record time-stamped terminal-voltage and current measurements during selected charging, discharging, and vehicle-operation conditions. The subsystem was intended for electrical-data acquisition and logging; it was not designed to provide state-of-charge estimation, state-of-health assessment, battery modelling, parameter identification, capacity characterisation, or thermal monitoring. The hardware implementation of the circuit is presented in Figure 4, and its circuit schematic is presented in Figure 5.

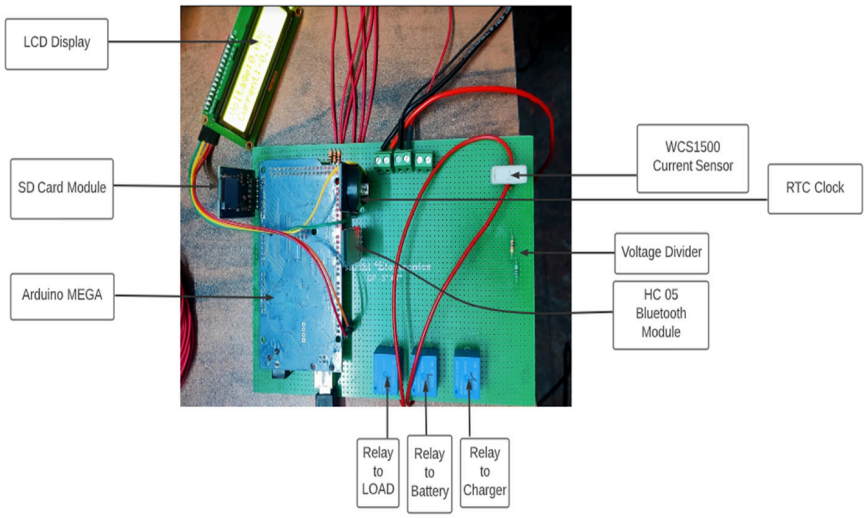


Figure 4. Hardware implementation and principal components of the real-time battery data logger prototype.

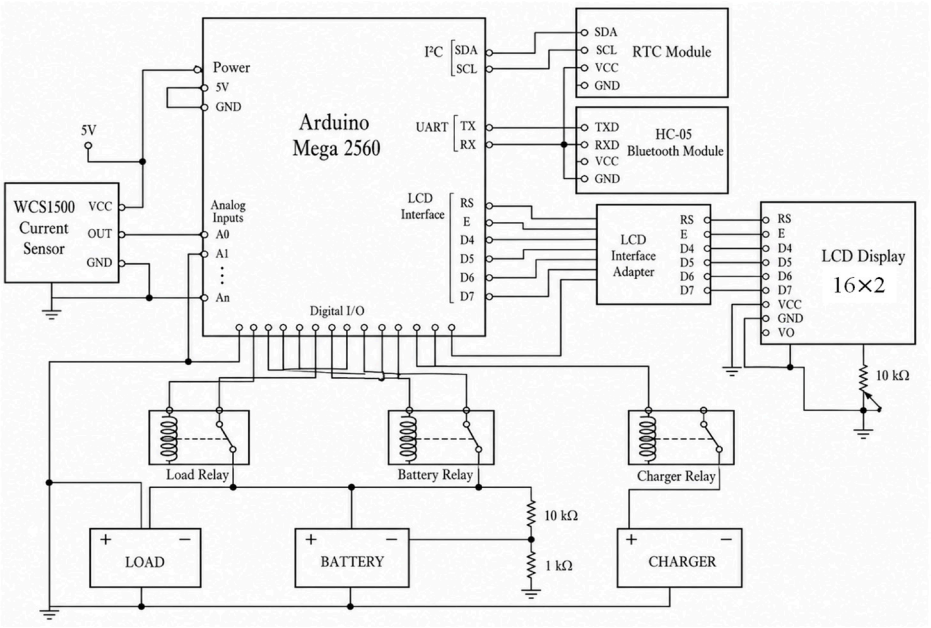


Figure 5. Functional connection diagram of the battery voltage–current data logger and relay-based operating-mode selection system [28].

The system includes an Arduino Mega controller, a WCS1500 current sensor, a voltage-divider circuit made of 10 kΩ and 1 kΩ resistors, a real-time clock (RTC) module, an SD card module, an HC-05 Bluetooth module, a liquid-crystal display (LCD), and three relay channels connected to the load, battery, and charger pathways. The RTC module is used to obtain a timestamp for the collected measurements, while the SD card module is used to store the voltage and current data for later analysis. The HC-05 module enables wireless data transmission, and the LCD displays the measured electrical parameters and operating status. The relay network enables charging, discharging, and direct charger-to-load operation. The voltage and current were measured at 1 s intervals.

A 10 kΩ/1 kΩ voltage divider was used to scale the battery voltage to a level suitable for the Arduino analogue input. At the maximum battery voltage of approximately 42 V, the divider output is approximately 3.82 V, and the divider current is approximately 3.82 mA. The total power dissipation is approximately 0.16 W, with most of the dissipation occurring in the 10 kΩ resistor. These values provide the required voltage scaling while maintaining a relatively low source impedance for stable analogue sampling. However, the divider forms a continuous current path across the battery when it remains connected. If continuously connected, it would consume approximately 0.092 Ah per day; therefore, this arrangement is suitable for the selected prototype tests but may be undesirable during prolonged storage. Future designs could reduce this quiescent current by using higher-value resistors with suitable signal conditioning or a switched voltage-measurement circuit. In addition, resistor tolerance and analogue-to-digital-converter characteristics can contribute to measurement uncertainty; these effects were not independently quantified in the present study.

2.4. Rider-Mass Based Range Estimation

The range-estimation approach was designed for low-speed electric kick-scooter use on nearly level roads within campuses and gated communities. Under these conditions, the effects of road gradient and aerodynamic drag were assumed to be small, while the electrical demand during acceleration and the rolling resistance were expected to depend on the rider's weight. The goal was not to create a high-fidelity vehicle-dynamics or electrochemical battery model but to compare the relative electrical demand and theoretically estimated travel range for different rider masses. Three separate pilot trials were conducted

for rider masses of 60 kg, 80 kg, and 140 kg. The same electric kick scooter and battery pack were used for all trials. The onboard data logger was used to record the battery voltage and current at 1 s intervals. The analysis was considered a preliminary comparison because repeated trial data and detailed information on the route, road surface, tyre pressure, initial state of charge, and ambient conditions were not available. The instantaneous electrical power was determined using Equation (3) [28,29].

$$P(t) = V(t)I(t) \tag{3}$$

where P(t) is the instantaneous electrical power and V(t) and I(t) are the measured battery voltage and current, respectively.

The nominal battery energy was calculated using Equation (4) [29,30].

$$E_{nom} = V_{nom}\, C_b \tag{4}$$

where E_{nom} is the nominal battery energy in Wh, V_{nom} is the nominal battery voltage and C_b is the rated battery capacity in Ah.

The battery-energy term in Equation (4) represents nominal battery energy rather than directly measured usable energy. In practical operation, the usable energy available before BMS cutoff may be lower because the permissible voltage and SOC window are restricted to protect the battery pack. Usable capacity can also be affected by discharge rate, battery temperature, ageing, and terminal-voltage behaviour. Therefore E_{nom} is used only for the present theoretical comparison of the pilot cases and should not be interpreted as the actual energy available for vehicle operation. The average electrical power over the measurement period was determined from Equation (5) [30,31].

$$\overline{P} = \frac{1}{N}\sum_{k=1}^{N} V_k I_k \tag{5}$$

where N is the number of recorded samples, and V_k and I_k are the measured voltage and current at the kth sampling instant.

When only average current was available, it was approximated as in Equation (6) [29–31].

$$\overline{P} \approx V_{nom}\, \overline{I} \tag{6}$$

The estimated operating duration and travel range were calculated using Equations (7) and (8) [29–31].

$$t_{est} = \frac{E_{nom}}{\overline{P}} \tag{7}$$

$$R_{est} = \overline{v} t_{est} \tag{8}$$

where $\overline{v}$ is the assumed average scooter speed of 15 km/h, t_{est} is the estimated operation duration in hours, and R_{est} is the estimated travel range in kilometres. The calculated duration and range are theoretical ideal values because they were derived from the nominal battery capacity of 4.4 Ah, whereas the usable capacity may be lower because the battery management system restricts the permissible voltage and state-of-charge window. The calculation also does not account for drivetrain losses, discharge-rate effects, battery ageing, temperature and terminal-voltage variations, road gradient, or changes in acceleration. Accordingly, it was used only for a preliminary comparison of the pilot rider-mass cases and should not be interpreted as the actual travel range under real operating conditions.

2.5. Mobile Application Interface

The prototype of a mobile application was created to enable user authentication, identify available scooters in the vicinity, view vehicle status, allow the optional reservation of a scooter, unlock the scooter using a QR code, and monitor and terminate the ride.

Figure 6 illustrates the monitoring and selected control workflow of the mobile application prototype evaluated in this study. After the application is opened, a Bluetooth connection to the prototype scooter is attempted. Following a successful connection, the monitoring dashboard displays vehicle speed, battery terminal voltage, battery current, Bluetooth connection status, and selected operating-mode and system-status information. Where applicable, the user can select an available scooter-control function, and the corresponding command is transmitted through Bluetooth to the IoT control and communication board. The user can then continue monitoring, select control interactions or end the prototype demonstration. The workflow represents a prototype interface for monitoring and interaction; it does not represent a deployed fleet-management application and does not include SOC estimation or real-time range prediction.

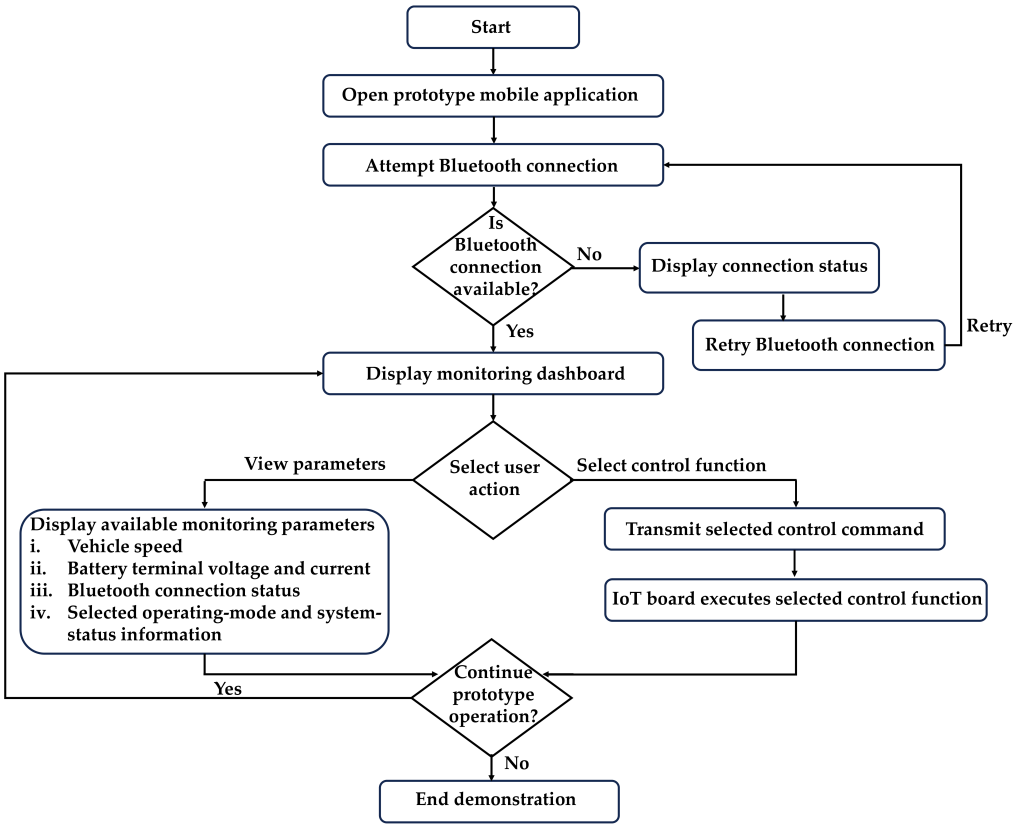


Figure 6. Operational workflow of the mobile application for electric kick-scooter access and ride monitoring.

3. Results

3.1. BLDC Motor Controller Performance

3.1.1. Simulated Switching-Signal Verification

The simulated gate states of the six metal–oxide–semiconductor field-effect transistors (MOSFETs) in the three-phase inverter are shown in Figure 7 and are labelled A1, A2, B1, B2, C1, and C2. The signals switch according to the six-step switching sequence (Table 2) determined by the Hall state. One upper-side and one lower-side MOSFET from different inverter legs are switched on during each commutation interval, while the third motor phase remains electrically floating. The upper- and lower-side switching states of each

inverter leg do not overlap, minimising the risk of simultaneous conduction within the same leg. The six signals have consistent relative phase shifts that are characteristic of 120° trapezoidal commutation. These results confirm the intended switching sequence at the simulation level before its implementation in the controller hardware.

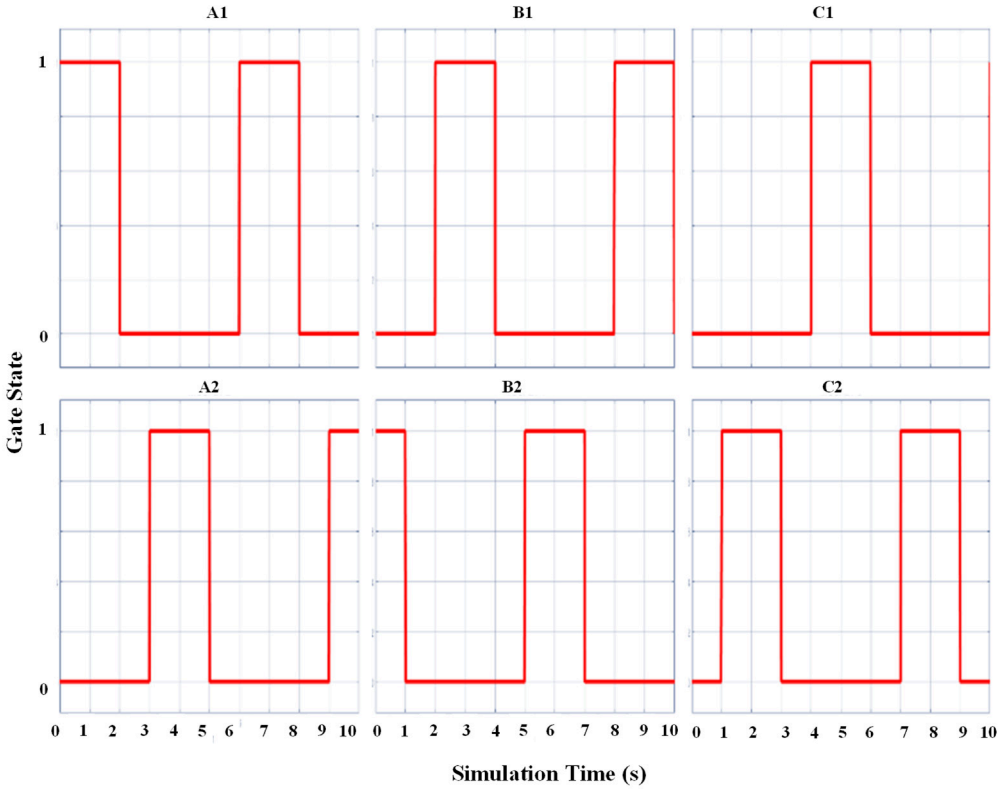


Figure 7. Simulated gate states for the six inverter MOSFETs during 120° trapezoidal commutation: A1, B1, and C1 represent the upper-side switches, while A2, B2, and C2 represent the corresponding lower-side switches.

3.1.2. Phase-Voltage Response

The simulated phase-terminal voltages are presented in Figure 8. The six-step inverter switching sequence produces positive, zero, and negative values for each voltage. The three phase-terminal voltages vary sequentially, as expected from a three-phase inverter operating under 120° trapezoidal commutation. These results confirm the switching logic and voltage-state sequence at the simulation level. The model excludes the quantification of hardware switching losses, voltage ripple, electromagnetic torque ripple, controller efficiency, and the dynamic response of the BLDC motor.

3.1.3. Hardware Implementation and Speed Testing

This Hall-sensor-based prototype controller was implemented using a microcontroller, gate-driver circuits, and a six-MOSFET three-phase inverter. The prototype was then tested with the 350 W BLDC hub motor using a digital tachometer. A maximum rotational speed of 1753 rpm was achieved during the preliminary test. The measurements demonstrate successful motor operation using the adopted six-step commutation scheme. However, more controlled testing under varying load conditions is required to evaluate speed regulation, controller efficiency, torque output, switching behaviour, and thermal performance. The simulated duty cycles of the gate states and the phase-terminal voltages are consistent with the 120° trapezoidal commutation principle, which is used in low-cost BLDC drives because of its simple implementation and lower computational demand compared with more advanced methods such as FOC [22]. In the present work, the electrically assisted

kick-scooter studies of Corno et al. [8,9], which focused mainly on kick detection and closed-loop velocity control, were extended through the implementation of the BLDC motor controller, along with battery monitoring and fleet-orientated functions. The findings reported in this study complement those of Asperti et al. [10], which were obtained from a fixed test-bench setup that did not consider vertical dynamics or rider comfort.

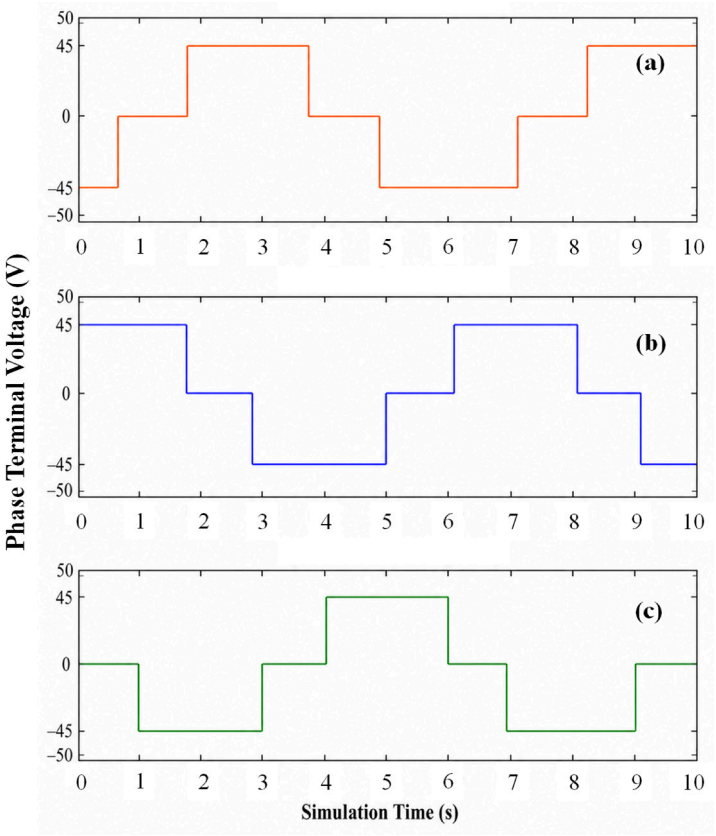


Figure 8. Simulated three-phase terminal voltages generated by the 120° trapezoidal commutation sequence: (**a**) Phase A terminal voltage, VA; (**b**) Phase B terminal voltage, VB; and (**c**) Phase C terminal voltage, VC.

3.2. Battery Data Logger Operation and Mode Selection

Battery voltage and current during charging, discharging, and load operation using the charger were measured by the integrated data logger. The required electrical path between the battery, charger, and load was selected using three relay channels controlled by the Arduino Mega digital pins D22, D24, and D26. Table 3 summarises the programmed relay states for the three operating conditions. If the measured battery voltage was 30 V or less, the relay combination for charging mode was activated. At a battery voltage of 42 V or greater, the relay states were switched to allow the battery to discharge through the load. When the load was electrically connected directly to the charger, a different relay combination was used. In all operating modes, the data logger continuously measured the battery voltage and current for further analysis of the charging and discharging behaviour.

Table 3. Relay states and selected operating modes for charging, battery-to-load discharging, and direct charger-to-load operation.

Operating Condition	D22	D24	D26	Selected Operating Mode
Battery voltage $V \leq 30$ V	ON	ON	OFF	Charging mode
Battery voltage $V \geq 42$ V	OFF	ON	ON	Battery-to-load discharging mode
Charger-supplied load operation	ON	OFF	ON	Direct charger-to-load mode

3.3. Battery Charging and Discharging Characteristics

During the charging and discharging tests, the battery voltage and current were recorded continuously at a sampling interval of 1 s. The corresponding voltage and current profiles are shown in Figure 9. The battery voltage gradually increased from 30 to 35 V during the charging test, while the charging current decreased slightly from 1.80 to 1.73 A. The short-term variations in the recorded signals could be due to measurement noise and fluctuations in the charging supply.

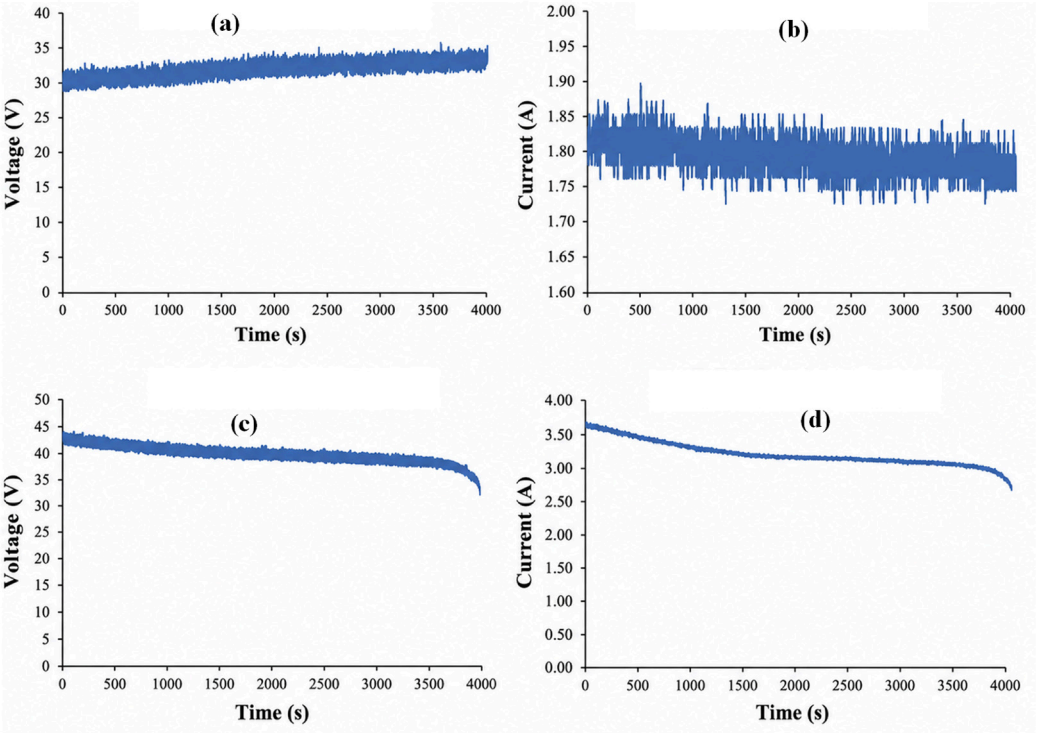


Figure 9. Battery voltage and current profiles recorded during charging and discharging tests: (**a**) charging voltage; (**b**) charging current; (**c**) discharging voltage; and (**d**) discharging current.

The test corresponds to a partial charging interval rather than a complete constant-current–constant-voltage charging cycle. However, the results show that the prototype data logger successfully recorded the gradual changes in battery voltage and current during charging. For the discharging test, a 12 Ω DC load was connected to the battery, and the battery voltage was monitored for about 3640 s. The terminal voltage dropped from approximately 43 V to 32 V, while the discharge current decreased from approximately 3.6 to 2.7 A. The proportional decrease in voltage and current indicates discharge through an approximately constant resistance. During most of the test, the voltage decreased gradually and then dropped more sharply near the end of the test period. The results demonstrate the ability of the developed data-logging system to capture battery behaviour during charging and discharging over time.

The charging and discharging tests were conducted to demonstrate the functional operation of the developed battery voltage–current data logger and to observe terminal-voltage and current responses under selected conditions. They were not intended to provide complete battery characterisation or a direct measurement of usable battery capacity. The charging profile represented only a partial interval rather than a complete constant-current–constant-voltage cycle, while the resistive-load discharge test was not a controlled full discharge-capacity evaluation. Full charge–discharge cycling, capacity evaluation over controlled discharge rates, temperature-dependent testing, internal-resistance analysis, and electrochemical characterisation were outside the scope of the present study. Therefore,

the results should be interpreted as evidence of logger operation and recorded electrical trends rather than a comprehensive assessment of battery performance. The measurement accuracy of the prototype logger was not independently validated against calibrated reference instruments. Sensor calibration, measurement uncertainty, repeatability, and data-loss rate were not quantified. Therefore, the recorded values are interpreted primarily as terminal-voltage and current trends demonstrating functional data acquisition, rather than as independently validated measurements of battery performance.

3.4. Preliminary Rider-Mass Comparison of Current Consumption and Theoretical Range Estimates

The rider-mass investigation comprised one pilot trial for each of the 60, 80, and 140 kg cases. The average recorded battery currents were 3.88, 7.40, and 8.30 A, respectively. The corresponding average power, theoretical operating duration, and theoretical range estimates are presented in Table 4. Although the recorded values differed across the three pilot cases, the limited number of cases and absence of repeated trials do not permit statistical analysis or causal interpretation. The results are therefore presented as descriptive preliminary observations only. The theoretical operating duration and range values were calculated using nominal battery energy, the average current recorded during each pilot trial, and an assumed constant speed of 15 km/h. The calculated theoretical ranges were 17.01, 8.92, and 7.95 km for the 60, 80, and 140 kg cases, respectively. These values are calculated outputs of the energy-based method and are not directly measured travel distances. The calculation does not account for usable battery capacity or battery management system operating limits, drivetrain efficiency, battery ageing, temperature effects, road gradient, acceleration variability, tyre pressure, wind, or terminal-voltage variation. Therefore, the reported values should not be interpreted as validated predictions of actual scooter range. Future validation requires repeated controlled trials for each rider-mass case, direct travel-distance measurements, usable-capacity testing, and consideration of relevant vehicle, route, battery, and environmental variables.

Table 4. Theoretical energy-based operating-duration and range estimates for the three rider-mass pilot trials.

Rider Mass (kg)	Average Current (A)	Average Power (W)	Estimated Duration (min)	Estimated Range (km)
60	3.88	139.68	68.04	17.01
80	7.40	266.40	35.68	8.92
140	8.30	298.80	31.81	7.95

The present analysis used average electrical values recorded during each single-run pilot trial. Time-resolved voltage, current, and power plots for the individual rider-mass cases were not included. Future work should use repeated controlled rider-mass trials with time-resolved electrical measurements to characterise transient energy demand and improve the robustness of the range-assessment approach.

3.5. Mobile Application Prototype Demonstration

The mobile application prototype offers an interface for accessing the scooters and monitoring their operation. The application's landing screen and monitoring dashboard are illustrated in Figure 10. The dashboard displays trip distance, vehicle speed, battery voltage, battery current, battery temperature, Bluetooth connection status, and vehicle-control functions such as lighting, riding mode, power, and emergency controls. The prototype represents the intended interaction between the rider and the electric kick scooter

for campus, industrial, and gated-community applications. No quantitative assessment of usability, communication latency, or application response time was conducted.



(**a**) (**b**)

Figure 10. Mobile application prototype for electric kick-scooter operation: (**a**) application landing screen and (**b**) vehicle monitoring dashboard.

The mobile application was implemented as a prototype interface to demonstrate the intended vehicle-status monitoring workflow and selected user interactions. The displayed functions included vehicle speed, battery terminal voltage and current, Bluetooth connection status, and selected scooter-control options. The application was not evaluated as a fully deployed fleet-management system. In particular, quantitative usability assessment, communication latency, long-term cloud operation, multi-scooter connectivity, and field deployment were outside the scope of the present study.

4. Discussion

The results demonstrate the practical implementation of the BLDC motor controller, battery data-logging system, rider-mass-based range-estimation approach and mobile application prototype. The simulated gate states and phase-terminal voltage patterns matched the intended six-step pattern for 120° trapezoidal commutation. Preliminary tachometer measurements were also performed to verify motor operation using the implemented controller. However, controller efficiency, torque output, switching losses, voltage ripple and thermal performance were not evaluated. Time-dependent voltage and current variations were measured during partial charging and resistive-load discharging. No uncertainty or calibration analyses were conducted, and these results only indicate the data-acquisition capability of the prototype and may not represent the accuracy or repeatability of the measurements. The rider-mass pilot study demonstrated a trend of higher mean current demand for the 60, 80 and 140 kg rider groups, with lower theoretical range estimates for the heavier rider groups. This trend is similar to that reported in previous studies on electric vehicles and micromobility, in which electrical demand varied with payload and operating conditions [15–18]. However, the results are based on only three individual trials and idealised calculations using nominal battery energy and an assumed average

speed. As such, they do not represent a statistically valid range model but rather a preliminary trend. Kick detection and velocity control have been the focus of most previous studies on electric kick scooters [8,9], along with rider comfort and vertical dynamics [10]. This study, however, considers the system-level integration of motor control, battery voltage/current logging, preliminary range estimation and a mobile monitoring interface. The main contribution is the implementation of these functions within a single prototype, rather than a comparison with commercial controllers or the development of a fully validated fleet-management system.

4.1. Comparison with Commercial Electric Kick-Scooter Platforms

Table 5 positions the developed prototype relative to representative commercial electric kick-scooter platforms. Commercial scooters generally incorporate proprietary integrated controllers, manufacturer-specific battery-management functions, and mature mobile-app ecosystems. In contrast, the present work is a preliminary research prototype intended to demonstrate a Hall-sensor-based BLDC controller, battery voltage–current logging, a limited rider-mass energy comparison, and a prototype mobile interface. The comparison is feature-based and does not represent a direct performance benchmark. Commercial range values are manufacturer-reported under specified test conditions, whereas the ranges reported in the present study are theoretical estimates derived from nominal battery energy, average current from three pilot trials, and an assumed constant speed. Therefore, these values are not directly comparable.

Table 5. Feature-level comparison of the proposed prototype with representative commercial electric kick-scooter platforms.

Feature	Xiaomi Electric Scooter 4 Ultra [32]	Segway Ninebot MAX G2 (Canadian Specification) [33]	Proposed Platform
Platform purpose	Commercial electric scooter	Commercial electric scooter	Preliminary research prototype
Motor/drive specification	500 W rated brushless motor; 940 W maximum power	450 W rated rear-wheel-drive motor; 1000 W maximum power	350 W, 6-inch BLDC hub motor
Drive-control architecture	Detailed commutation and controller architecture not disclosed in the cited manufacturer specifications	Detailed commutation and controller architecture not disclosed in the cited manufacturer specifications	Hall-sensor-based six-step BLDC control with 120° trapezoidal commutation
Battery and energy-management information	Lithium-ion battery, 561.5 Wh; approximately 6.5 h charging time	36 V, 15.3 Ah lithium-ion battery, 551 Wh; BMS protection functions; approximately 5 h charging time	36 V, 4.4 Ah Li-ion battery with BMS; time-stamped voltage–current logging only
Published range-test condition	Fully charged battery, 75 kg load, 15 km/h, flat road	Fully charged battery, 75 kg load, 25 °C, average 16 km/h on pavement	Pilot trials at 60, 80, and 140 kg
Range information	Up to 70 km, manufacturer-reported *	Up to 43 miles (70 km), manufacturer-reported theoretical range *	Theoretical energy-based range estimates for the three rider-mass conditions
Mobile interface	Xiaomi Home/Mi Home app for ride data, remaining-range display, and fault notifications	Segway-Ninebot Bluetooth mobile-app support	Prototype mobile interface; no fleet deployment, latency, or usability evaluation

Table 5. *Cont.*

Feature	Xiaomi Electric Scooter 4 Ultra [32]	Segway Ninebot MAX G2 (Canadian Specification) [33]	Proposed Platform
Evaluation context	Manufacturer-reported specifications; not independently verified in this study	Manufacturer-reported specifications; not independently verified in this study	Subsystem tests and pilot rider trials

* Manufacturer-reported range values were obtained under different specified test protocols and were not independently verified in the present study.

4.2. Limitations and Future Work

The battery subsystem was limited to terminal-voltage and current acquisition and time-stamped logging. The charging and resistive-load discharging tests demonstrated logger operation and the electrical response under selected conditions; they did not provide complete battery characterisation. No validated state-of-charge or state-of-health estimation, battery-model development, parameter identification, capacity testing, or thermal measurement and analysis was performed. A comprehensive battery assessment would require calibrated full charge–discharge testing, repeated measurements under controlled temperature conditions, comparison with reference instruments, and validated battery-state estimation and modelling methods. The rider-mass assessment was preliminary, as only one pilot trial was conducted for each of the 60, 80, and 140 kg cases. The reported ranges were theoretical estimates calculated from nominal battery energy, average current, and an assumed constant speed of 15 km/h; they were not directly measured travel distances. The calculation did not include usable battery capacity, including battery management system operating limits, drivetrain efficiency, or other vehicle, route, battery, and environmental conditions. Future validation requires repeated controlled trials across rider-mass cases, direct travel-distance measurements, usable-capacity testing, and inclusion of these variables in an improved range-prediction model. In addition, calibration against reference instruments, measurement uncertainty, repeatability, and data-loss rate were not independently evaluated; these are required for future validation of logger accuracy and reliability. Detailed voltage, current, and power time-series analysis was not performed for the single-run rider-mass pilot trials; repeated time-resolved measurements are required for a more robust assessment of transient energy consumption.

5. Conclusions

The design, implementation, and preliminary evaluation of an electric kick-scooter platform comprising a Hall-sensor-based BLDC motor controller, battery voltage–current data logger, preliminary rider-mass-based theoretical range calculation, and mobile application prototype were presented. The simulated gate states and phase-terminal voltage responses were consistent with the intended six-step sequence for 120° trapezoidal commutation. Preliminary hardware testing recorded motor rotational speeds between 1632 and 1753 rpm, confirming basic motor operation with the implemented controller. During partial charging and resistive-load discharging, the battery voltage–current data logger continuously recorded terminal-voltage and current patterns. The recorded voltage increased from 30 to 35 V during charging and decreased from 43 to 32 V during discharging over 3829 and 3640 s, respectively. These tests demonstrate logger operation under the selected conditions but do not constitute complete battery characterisation or independently validated measurement accuracy.

The three rider-mass pilot trials recorded average currents of 3.88, 7.40, and 8.30 A for the 60, 80, and 140 kg cases, respectively. When these values were used with nominal

battery energy and an assumed constant speed of 15 km/h, the corresponding theoretical range estimates were 17.01, 8.92, and 7.95 km. These values are calculated outputs of the energy-based method rather than directly measured travel ranges. Therefore, they provide descriptive pilot observations only and do not establish a validated, causal, or predictive relationship between rider mass and actual scooter range. The mobile application was presented as a prototype interface for intended vehicle-status display and selected control functions; it was not evaluated as a fully deployed fleet-management application. Overall, this study provides a preliminary prototype-level basis for further development and controlled evaluation of electric kick-scooter drive control, voltage–current logging, range assessment, and mobile-interface functions.

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Abbreviations

B2B	Business-to-Business
BEMF	Back-Electromotive Force
BLDC	Brushless Direct Current
BMS	Battery Management System
CAN	Controller Area Network
DC	Direct Current
EV	Electric Vehicle
FOC	Field-Orientated Control
GPS	Global Positioning System
IoT	Internet of Things
LCD	Liquid-Crystal Display
MOSFET	Metal–Oxide–Semiconductor Field-Effect Transistor
OLED	Organic Light-Emitting Diode
PWM	Pulse-Width Modulation
QR	Quick Response
RTC	Real-Time Clock
SD	Secure Digital
SOC	State of Charge

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